

# Connecting scales in reaction engineering

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**As part of the March Focus issue of *Nature Chemical Engineering*, we asked 13 leading researchers to spotlight a challenge or opportunity in reaction engineering that they believe holds particular promise for advancing this core area of chemical engineering research and practice.**

Reaction engineering necessitates a detailed understanding and multivariable optimization of processes across multiple length and time scales, ranging from molecular-level reaction kinetics and reactor-scale transport phenomena to system-level integration and scale-up. This multiscale approach is crucial for transforming chemical processes from ideation to commercialization and will be particularly important in accelerating bench-to-market timelines. In this Viewpoint, we have asked 13 members of the community to share their perspectives on the most exciting opportunities that lie ahead for their respective topics in reaction engineering. Here is what they said.

## Jeremy Luterbacher: the vexing (but vital) challenge of sustainable transition

Much of the content of today's advanced reaction engineering classes is based on research done between the 1930s and 1970s, largely in response to the rapid expansion of petrochemical processing. We all learned the names of Thiele, Weisz and Boudart who explained the effects of transport in porous catalysts. Thiele (at Standard Oil) and Weisz (at Mobil Oil) even made their contributions while working for petroleum companies.

Given the inevitable transition to chemical manufacturing from more sustainable feedstocks such as CO<sub>2</sub>, plants or mixed plastics, another boom in chemical reaction engineering is likely on the horizon. Lignin, a plant fraction that is by far nature's largest source of aromatic moieties, is a perfect example of an attractive sustainable source for chemicals that requires new advances in reaction engineering to exploit effectively.

Lignin's macromolecular nature requires harsh liquid-phase processing, while its



oxygenated functionalities, especially in activated benzylic positions, makes it prone to uncontrolled condensation reaction. Such properties will require processes of unprecedented complexity coupled with accurate three-dimensional models that capture its transport from plant to catalyst. Its nature as a heteropolymer also means that understanding lignin depolymerization with simple kinetics is out of the question. Lignin likely needs to be represented as a stochastic set of differing polymers, with its transformation captured by kinetic Monte Carlo approaches.

Future chemical engineering curricula will hopefully reflect on this coming period as a time when reaction engineers again rose to society's challenges – this time imposed by the global sustainable transition.

## Bert Weckhuysen: probing and steering reactors with all-in-one catalyst particles

Future chemicals, fuels and materials should be sustainably produced from more renewable feedstocks, such as agricultural and municipal waste, containing different impurities. These new feedstocks together with the use of wind and solar energy will require more compact, flexible and intermittently operating chemical reactors.

I envisage in the coming decades an entire new generation of 'intelligent' catalyst materials that not only probe what is happening within these reactors but also actively control and steer the chemical processes taking place. These probing and steering catalyst particles would revolutionize the way we operate industrial processes. The first developments are

under way: particles not only catalyze chemical reactions, but also can serve as probes of the local catalytic chemistry and reaction temperature with operando spectroscopy. Similar efforts are on their way to locally probe reaction pressures.

The next steps are to use this information to steer a catalytic process by locally triggering or suppressing chemical reactions, including cleansing and hence rejuvenating a catalytic surface covered by, for example, coke deposits or poisons, optically heating or cooling a reactor bed, thereby suppressing unwanted side and runaway reactions, and actively responding to sudden feedstock composition changes or intermittencies in impurities.

Coupling this wealth of information – gathered by novel 'all-in-one' catalyst particles positioned within a reactor bed – with artificial intelligence tools makes it possible to not only test fundamental concepts in catalysis but also operate chemical reactors flexibly, robustly and safely.

## Sophia Haussener: coupling photons into a chemical reactor

Solar energy conversion processes and reactors will play a crucial role in transitioning toward a sustainable fuel economy and chemical industry. In the future, rather than extracting and refining fossil fuels, we will harness and convert solar energy. However, designing, implementing, characterizing and optimizing such systems presents a unique challenge compared with traditional, non-solar processes: photons must be effectively coupled into the system. This requires (windowed) reactor openings and specialized absorbing structures. These structures often serve multiple functions, acting as both absorbers and reaction interfaces.

Beyond photon absorption, additional multiphysical phenomena – such as heat and mass transfer – occur simultaneously, requiring careful coordination to ensure balanced photon, charge and mass fluxes. These complex processes operate across multiple scales, influenced by surface characteristics, interfaces, boundary layers and mesostructures. A key challenge lies in understanding

and co-designing the nonlinear interactions of these multiscale and multiphysical processes to develop functional and feasible solar processes and reactors. Once constructed, these systems must undergo rigorous testing and characterization (initially under well-controlled conditions), necessitating advanced characterization procedures. Ultimately, pilot-scale demonstrations are essential for collecting real-world data, assessing practical feasibility, developing scalable approaches and analyzing the dynamic response of the system to intermittent solar irradiation.

## Beatriz Roldan Cuenya: no performance without fluctuation

Interfacial catalysis functions at the molecular level through active sites enabling the execution of the rate-controlling step. These sites are (re)generated by surface dynamics. Joint action of temperature or applied electrical potential and the local adsorbate coverage define the local chemical potential and enable surface/subsurface chemistry driving dynamics. Nuclei form, the growth of which is inhibited by the presence of adsorbates. Adequate timing of adsorption, diffusion and desorption ensure the persistence of the reactive surface. The interface deactivates when the stable phase grows from its nuclei, which are the active sites.

These atomistic processes are coupled to mesoscopic and macroscopic dimensions in space and time through the local chemical potential given by global reaction conditions, mass and energy transport phenomena, and their instabilities. The best performance of the reactor–catalyst–reagent system is achieved when the atomistic time constants of surface dynamics and the chemical reaction coincide, resulting in a steady state. Interference from meso- and macroscopic transport instabilities must be avoided. Resulting gradients in local chemical potential induce growth of nuclei and initiate the deactivation of the system. A combined catalyst–reactor–process design, including space–time–resolved feedback control of the process variables, will enable future engineering exploitation of the advanced insights into atomistic reaction details.

## Daniel E. Resasco: harnessing the complexity of multiphase catalysis

Interfacial dynamics in multiphase catalysis are crucial for advancing sustainable energy and chemical manufacturing. These systems, defined by complex interactions at gas–liquid–solid or liquid–liquid–solid interfaces, determine catalytic behavior, reaction rates



and selectivity. Interfacial environments generate localized gradients, electric fields and polarity shifts that stabilize intermediates and enable challenging reactions.

Water exemplifies this complexity, acting as a solvent or reactant to stabilize intermediates, polarize bonds through hydrogen-bonded networks and influence catalytic outcomes. Solvent effects and non-idealities, such as activity coefficients and selective solvation, also play an important role in reaction kinetics, highlighting the need for tailored solvent systems.

At the active site, multiple phenomena coexist and interact with the surroundings, but identifying which factors affect reaction rates is essential. For instance, is the rate determined by transition state energy, local reactant concentration or chemical potential in non-ideal systems? Can thermodynamic excess functions be estimated to predict rate changes? Studies show that distances between charged species and active sites, along with surface density of active sites, influence rates via local electric fields and bond polarization.

Multiphase phenomena introduce additional challenges, including mass transfer limitations and phase-specific gradients. Addressing these requires advanced tools such as in situ characterization, multiscale modeling and machine learning. Future efforts must focus on mapping interfacial dynamics, understanding non-ideal behavior and designing optimized systems. These advancements will provide transformative solutions for renewable energy and sustainability, driving interdisciplinary innovation in chemical engineering and beyond.

## Carlos G. Morales-Guio: bridging reaction kinetics and scale-up

Although electrolyzers for hydrogen production and the chlor-alkali process operate at commercial scales, scaling electrochemical reactors for chemical production from CO<sub>2</sub>, N<sub>2</sub>, CH<sub>4</sub> and dissolved species such as nitrates

or carbonates remains a challenge. This stems from the inherently complex nature of electrocatalytic transformations. Even at the bench scale, mass, heat and charge transport are intricately coupled with electrocatalytic kinetics, dictating reaction selectivity and efficiency. Maintaining catalyst performance when scaling electrolyzers from a few square centimetres of electrode area in lab-scale cells to the square-metre-scale required for industrial implementation is already difficult, even when keeping the same reactor format. Compounding this, the optimal industrial electrolyzer design for many of these transformations remains uncertain.

Critically, what we measure experimentally is reactor kinetics, not intrinsic reaction kinetics. The same catalyst can exhibit different selectivities in different reactor designs if local mass, heat and charge transport conditions vary. This makes it difficult to decouple the fundamental catalyst kinetics from transport effects, complicating reactor design and scale-up.

A viable path forward involves acquiring large catalytic datasets spanning a broad range of temperature, pressure, reactant concentration, hydrodynamics and surface potentials to develop physics-informed models that bridge molecular-scale catalyst kinetics with reactor-scale performance. By integrating multiphysics simulations, machine learning, automation and model predictive control, future electrolyzers can be designed to meet stringent performance metrics at scale, advancing reaction engineering for sustainable chemical production.

## Feng Jiao: engineering toward commercial scale

CO<sub>2</sub> electrolysis offers immense promise for sustainable chemical and fuel production, but several challenges must be overcome for commercial deployment. First, enhancing electrode durability, particularly for gas diffusion electrodes, is essential. Under prolonged operation, gas diffusion electrodes are prone to flooding and degradation, which drastically reduces device lifetimes. There is a pressing need to develop advanced techniques that probe the dynamic behavior of catalysts and interfaces under industrially relevant conditions, providing crucial insights into failure mechanisms and guiding the design of more robust electrodes. Additionally, scaling up the electrolyzer requires careful attention to flow field design, mass transport and heat management.

Traditional laboratory cells often struggle to maintain efficiency as reactor sizes grow, largely due to transport limitations



and heat management. Integrating experimental efforts with computational transport modeling will accelerate the optimization of electrode geometry, channel structures and operating conditions to minimize losses. Finally, comprehensive process design and integration must guide technology development beyond the core electrochemical unit. Emphasizing the balance of plant, including gas handling, thermal management and product separation, can significantly influence overall energy consumption and cost. Techno-economic analysis should underpin these decisions, ensuring that every engineering trade-off is evaluated against both financial and sustainability metrics. By combining these efforts, CO<sub>2</sub> electrolysis can progress from laboratory-scale demonstrations to commercially viable systems that meaningfully contribute to a carbon-neutral future.

## Nanfeng Zheng: industrialization of water electrolyzers

Water electrolyzers play a pivotal role in industry by transforming renewable but fluctuating power into green hydrogen. Despite major progress in recent decades, the large-scale application of water electrolyzers for green hydrogen production remains in its early stages. Green hydrogen currently costs more than twice as much as grey hydrogen, even with low renewable electricity costs, mainly due to high capital costs, limited efficiency and high failure rates of electrolyzers. To accelerate the widespread adoption of green hydrogen in industrial applications, three key technical challenges must be addressed:

**Electrode degradation:** achieving high current densities is crucial for reducing electrolyzer capital costs, often requiring higher voltages and porous electrodes. However, increased mass transport resistance and voltage accelerate both chemical and physical degradation of the electrodes (for example, changes in composition and porous structure, cracking, ionomer migration). Further research is needed to better understand degradation under harsh conditions.

**Integration with fluctuating electricity and downstream processes:** while hydrogen crossover must be managed to prevent safety hazards during dynamic operations, the hydrogen pressure generated by electrolyzers also needs to be optimized to meet the requirements of downstream processes, including oil refining, ammonia or methanol synthesis, and hydrogenation of fine chemicals.

**Reliability:** several key materials and components used in electrolyzers are not yet



mature enough for large-scale green hydrogen production, leading to high failure rates. To improve the overall system reliability, a deep understanding of how failures at the material level propagate to the system level is required.

## Kazunari Domen: large-scale photocatalytic green H<sub>2</sub> production

Photocatalytic water splitting is being actively investigated as a practical means of producing green hydrogen on a large scale and at low cost<sup>1</sup>. Undoubtedly, a dramatic improvement in the solar energy conversion efficiency and the establishment of mass production processes for efficient photocatalysts are essential for the practical application of this technology. Meanwhile, given the low prices of the primary products, hydrogen and oxygen, reducing the production costs of reactors and reaction systems is an economically important issue. They should be much cheaper than existing chemical reactors.

Improving the energy efficiency and safety of the process for separating and recovering hydrogen from the raw product, wet oxygen-hydrogen gas, is also necessary to reduce the operating costs of reaction systems and to minimize the risk of industrial accidents. Another important area for the utilization of green hydrogen in the chemical industry will be the development of catalytic reactions and reaction systems that can operate at low temperatures and pressures, the conditions under which green hydrogen will be produced. There are many groups working on photocatalytic water-splitting materials at the laboratory level, but more research and development should be devoted to large-scale deployment from a reaction engineering perspective to make this technology truly useful.

## Patricia Concepción: hotspots in heterogeneous catalysis

In the case of highly exothermic reactions, efficient heat management is critical to prevent

the generation of hotspots (regions of locally elevated temperature) and in extreme cases uncontrollable temperature runaway. The generation of hotspots in heterogeneous catalysis can greatly affect the catalytic performance, leading to an overestimation of the intrinsic activity and a decrease in the stability of the catalyst.

There are different strategies to prevent hotspot formation. For example, in fixed-bed reactors, heat-dissipating elements can be incorporated to minimize local temperature excursions, such as by diluting the catalyst with thermally conductive SiC. The ratio of catalyst-to-diluent is typically determined experimentally. Additionally, controlling the reactant flow rate is critical, as mass transport limitations can create non-uniformity in the reaction bed. Another strategy to prevent hotspots is the use of fluidized beds or microchannel reactors to improve heat and mass transfer. Depositing the catalyst on, for example, ceramic monoliths favors heat dissipation, avoiding detrimental temperature gradients through increased surface area for heat exchange and a higher dispersion of the catalyst over the structure.

Finally, there are several tools to identify hotspots, such as temperature sensors located inside the catalyst bed, spectroscopic techniques (such as infrared thermography) and indirect methods such as deviations in the Arrhenius plot.

## Louise Olsson: stability and durability in heterogeneous catalysis

Stability and durability in heterogeneous catalysis is a key area for the scale-up of new innovative catalytic processes. Catalyst deactivation occurs for many reasons, but here the focus is on poisoning and coking. To ensure that relevant deactivation modes are studied, an important future path is to characterize well-defined catalysts that operate in industrial reactors. The learnings from those experiments can be used to design fundamental model studies.

The transition to a sustainable society requires the usage of bio-based feedstocks, especially from waste. However, bio-oils produced from these feedstocks are very challenging to work with as they can contain hundreds of different components, some of them that easily form char and coke, and inorganic poisons. One important future path is to pre-treat these bio-oils to remove inorganic poisons. Fixed-bed reactors offer a cost-effective solution for biofuel production through catalytic hydrogenation; however,

bio-oils are challenging to upgrade directly in these systems due to substantial char formation that plugs the reactor. Future research should target cost-efficient pre-treatment processes that stabilize the oils by converting the charring components, for example, through catalytic hydrodeoxygenation.

Multiscale reaction engineering is essential for the advancement of sustainable chemical processes. To accomplish this, it is imperative to establish rigorous, systematic testing protocols for stability and durability, thereby facilitating reproducibility and results comparison throughout the research community.

## Curtis Berlinguette: electrolytic decarbonization

The electrolytic generation of acid and base equivalents presents an opportunity to decarbonize heavy industry and the atmosphere. For example, electrolytic acid production can eliminate emissions from cement manufacturing by extracting calcium from limestone without a kiln<sup>2</sup>. Electrolytic acid can also potentially drive the release of CO<sub>2</sub> from carbon capture sorbents that would otherwise require heat or vacuum, and accelerate rock weathering to permanently store thousands of gigatonnes of CO<sub>2</sub> as carbonated minerals<sup>3</sup>.

Decarbonization electrolyzers, which integrate multiple process steps within a single reactor, must carefully manage mass transfer, bubbles, thermal regulation, current distribution and residence time distribution. These processes must all be performed while operating at applied potentials of ideally less than 3 V for the energy efficiency to be commercially viable.

The ion-conducting membrane ('separator') is often the primary source of voltage loss in an electrolyzer. The separator governs the transport and reactivity of acid and base within an electrolyzer and, therefore, impacts nearly all aspects of electrolyzer design and performance. However, there are few separators available that are suitable for industrial electrolysis. A broader effort from the research community is needed to help identify separators that are appropriate

for scaling decarbonization electrolyzers, or to identify electrolyzers that avoid the need for separators.

## Hongliang Xin: informed by theory, enhanced by data

Reaction engineering is inherently multiscale, shaped by the intricate interplay between molecular reaction kinetics, thermodynamics and reactor-scale transport phenomena. Traditionally, physics-based modeling has provided fundamental insights into these interdependent processes but often struggles to integrate heterogeneous data spanning multiple modalities, sources and scales. Today, artificial intelligence, particularly deep learning, is emerging as a promising tool to address these limitations. Data-driven technologies, when combined with physics-based simulations, offer a hybrid approach that significantly accelerates modeling of complex processes.

This data-enhanced multiscale modeling paradigm leverages high-performance computing infrastructure, large-scale datasets and cutting-edge simulation tools for process development and optimization. However, key challenges remain. The inherent opacity of deep learning models demands careful attention to model interpretability and uncertainty quantification. Overcoming these obstacles will require innovative strategies that embed domain theories or physical principles within algorithmic design, ensuring that models not only predict accurately but also provide understandable and actionable guidance – for example, for designing new processes to meet critical societal needs, ranging from sustainable chemical manufacturing to energy conversion and environmental remediation. As these hybrid frameworks mature, they promise to drive a new era of efficient, adaptive and semi-autonomous reaction engineering that will redefine the landscape of industrial practices, research and education.

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